Road Damage Detection

Literature Review & Evaluation

1. Literature Review

Published work on RDD2022 shows that one-stage YOLO detectors are the standard choice, that ensembles reach about 0.77 F1 on the full benchmark, and that small architectural additions give gains of one to two points. Our YOLO11n baseline (0.575 mAP@50) sits in the same family but uses no ensemble or custom modules.

1.1 Dataset

RDD2022 (Arya, Maeda, Ghosh, Toshniwal and Sekimoto, 2022) contains 47,420 road images from Japan, India, the Czech Republic, Norway, the United States and China, with more than 55,000 annotated damage instances in Pascal VOC format. The authors report that training on several countries improves generalization. Its four official classes are longitudinal crack, transverse crack, alligator crack and pothole. The Kaggle version we use adds a fifth class, Other Corruption, and its 38,385 images match the official labelled training set, re-split into 26,869 / 5,758 / 5,758 for train, validation and test.

1.2 Challenge results

The CRDDC'2022 challenge ran on RDD2022 with more than 50 participating teams. The winning team reached 0.770 F1 on the six-country leaderboard. Nine of the top 11 solutions used ensembles, and YOLOv5 and YOLOv7 were the most common base detectors. The organizers note that YOLO models tend to favor recall while Cascade R-CNN models favor precision, and that adding more models does not always raise accuracy.

1.3 Improved architectures

OBC-YOLOv8 modifies YOLOv8n with ODConv, a BoTNet attention block and Coordinate Attention. On a 4,378-image RDD-China subset it gains 1.8% mAP@50 over the baseline, at the cost of about 3.2M versus 3.0M parameters and a drop from 243 to 222 FPS. The authors also saw accuracy fall on Japan images because of weather and lighting, and shadows and manhole covers caused false detections.

1.4 Summary of related work

Study	Data	Method	Reported result	Relevance to our project
RDD2022	47,420 images, 6 countries	Dataset and benchmark	More than 55,000 labelled instances	Our training data
CRDDC'2022	RDD2022	Ensembles of YOLOv5/v7 and R-CNN with Swin backbone	Best F1 0.770 (six countries)	Shows the ceiling for ensembles and the value of augmentation
OBC-YOLOv8	RDD-China, 4,378 images	YOLOv8n + ODConv + BoTNet + Coordinate Attention	+1.8% mAP@50 over YOLOv8n	A lightweight upgrade path for our YOLO11n model

Study	Data	Method	Reported result	Relevance to our project
This project	RDD2022 split, 38,385 images	YOLO11n, 50 epochs, 640 px	0.575 mAP@50, 0.311 mAP@50–95 (validation)	Baseline to improve

1.5 Research gap

The cited studies report F1 or mAP on different subsets and splits, so their numbers cannot be compared directly with ours. Weak Pothole detection under class imbalance is the gap this project targets.

2. Feedback & Evaluation

The baseline detects Other Corruption and Alligator Crack well but misses most potholes. The table records our own assessment, and the last column is left for the lecturer's comments.

0.746	0.427	0.368
Best class mAP@50 (Other Corruption)	Worst class mAP@50 (Pothole)	Pothole recall

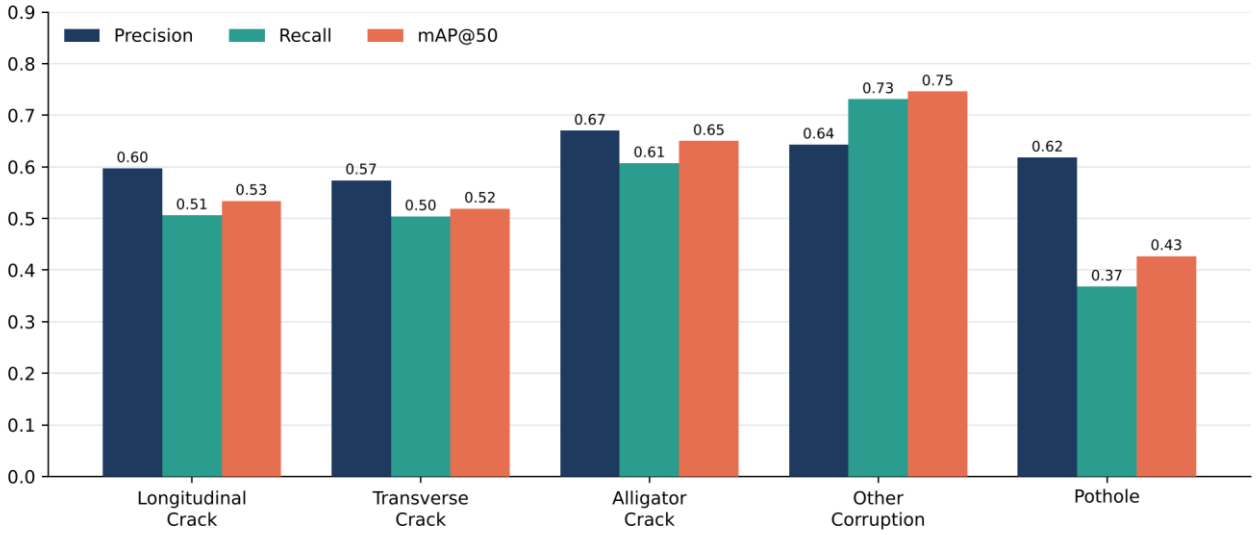


Figure 1 – Per-class precision, recall and mAP@50 on the validation split

Criterion	Evidence from the project	Lecturer feedback
Problem definition	Five damage classes on RDD2022; clear use case in road maintenance	
Data handling	38,385 images split 70/15/15; class balance and sample images inspected; 1,837 of 5,758 validation images contain no damage	
Model and training	YOLO11n, pretrained, 50 epochs, 640 px, batch 16, 6.15 h on Tesla T4	
Results	Precision 0.620, recall 0.543, mAP@50 0.575, mAP@50–95 0.311 (validation)	

Criterion	Evidence from the project	Lecturer feedback
Per-class analysis	Best: Other Corruption (0.746 mAP@50). Worst: Pothole (0.427 mAP@50, recall 0.368)	
Efficiency	2.58 M parameters, 6.4 GFLOPs, 4.3 ms inference per image	
Limitations	Metrics come from the validation split, which also picked best.pt; the test split was only inspected visually; no deployment yet	

3. Suggested Improvements

The first improvements to try target Pothole recall and the evaluation protocol, because they cost the least and address the weakest results.

Priority	Improvement	Why	Expected effect
1	**Evaluate the final model once on the test split**	Validation picked best.pt, so its numbers are slightly optimistic	Honest final metrics
2	**Oversample Pothole images and tune class loss weight**	Pothole is 10% of training objects and has recall 0.368	Higher Pothole recall
3	**Train a larger model (YOLO11s or YOLO11m)**	Nano model has only 2.58 M parameters	Higher mAP at a few ms more per image
4	**Raise input size to 800–1024 px**	Cracks are thin and small at 640 px	Better crack recall
5	**Add targeted augmentation (brightness, blur, shadows)**	Shadows and lighting caused errors in related work	Better robustness across countries
6	**Add attention modules as in OBC-YOLOv8**	Reported +1.8% mAP@50 on RDD-China	Small gain, slightly slower
7	**Ensemble or test-time augmentation**	Top CRDDC'2022 teams used ensembles	Higher accuracy, slower inference
8	**Tune the confidence threshold per class**	Default 0.25 may suppress weak Pothole detections	Better precision–recall balance
9	**Export to ONNX and build a demo app**	No deployment yet	Usable end-to-end system

4. Final Grading Criteria

The breakdown below is a proposal across four areas, totalling 100 marks.

Area	Marks	What is assessed
Documentation	25	Proposal, plan, literature review, risk log, final report, clear structure and references
Implementation	35	Data pipeline, YOLO11 training, improvements, working demo, code quality

Area	Marks	What is assessed
Testing & Evaluation	25	Metrics (mAP, precision, recall), per-class analysis, test-split results, comparison with the baseline
Presentation	15	Slides, live demo, answers to questions, teamwork

> Replace these weights with the official DEPI rubric once it is confirmed.